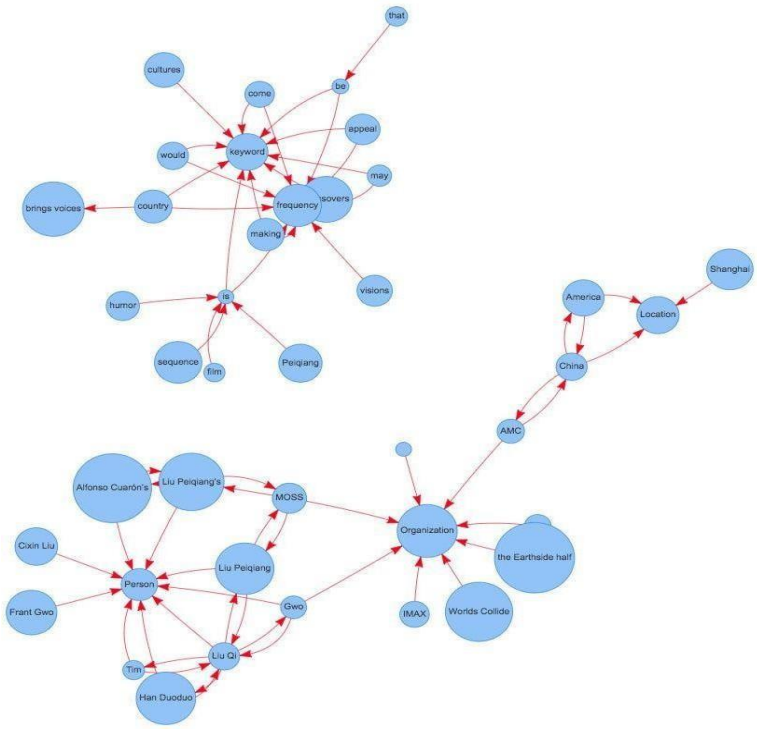
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** Representation of a document with their keywords using TextRank	
Experiment No: 09	**Date:**	**Enrolment No:92301733021**

Aim: Representation of a document with their keywords using TextRank

IDE: Google Colab

Theory:

TextRank is an algorithm based on PageRank, which often used in keyword extraction and text summarization. In this article, I will help you understand how TextRank works with a keyword extraction example and show the implementation by Python.



Understand PageRank

There are tons of articles talking about PageRank, so I just give a brief introduction to PageRank. This will help us understand TextRank later because it is based on PageRank. PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B. After we construct the whole graph, we can assign weights for web pages by the following formula.

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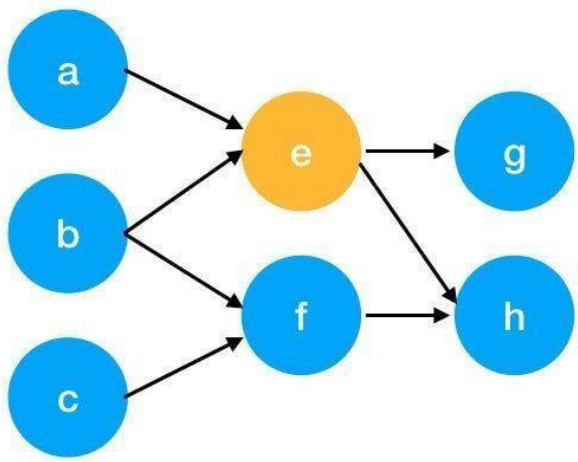
$$S(V_i) = (1-d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- S(Vi) - the weight of webpage i
- d - damping factor, in case of no outgoing links
- In(Vi) - inbound links of i, which is a set
- Out(Vj) - outgoing links of j, which is a set
- |Out(Vj)| - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above function to a simpler version.

$$In(v_e) = \{a, b\}, j \in \{a, b\}$$

$$\begin{aligned}\sum_{j \in \{a,b\}} \frac{1}{|Out(V_j)|} S(V_j) &= \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b) \\ &= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b) \\ &= S(V_a) + \frac{1}{2} S(V_b)\end{aligned}$$

We can get the weight of webpage e by the following function.

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$$S(V_e) = (1-d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

We can see the weight of the webpage e is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

Pre Lab Exercise:

1. **What are the key components of TextRank?**
   TextRank is based on graph representation where words or sentences are nodes and edges represent relationships between them. It uses co-occurrence of words and an iterative ranking algorithm (similar to PageRank) to identify important keywords or sentences.

2. **What are some limitations of TextRank for keyword extraction?**
   TextRank does not understand the meaning or context deeply, so it may miss important keywords. It also depends heavily on text structure and may not perform well on small or noisy datasets.

3. **What are the advantages of TextRank for keyword extraction?**
   TextRank is unsupervised and does not require labeled data. It is simple to implement and works well for extracting important keywords from large text efficiently.

Program (Code):

To be attached with

```
import nltk
import networkx as nx
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
import string
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
def textrank_keywords(text, window_size=4, top_n=5):

    words = word_tokenize(text.lower())
```

```
    stop_words = set(stopwords.words('english'))
    words = [w for w in words if w not in stop_words and w not in string.punctuation]

    graph = nx.Graph()

    for i, word in enumerate(words):
        for j in range(i+1, i+window_size):
            if j < len(words):
                graph.add_edge(word, words[j])

    scores = nx.pagerank(graph)

    ranked_words = sorted(scores.items(), key=lambda x: x[1], reverse=True)

    keywords = [word for word, score in ranked_words[:top_n]]

    return keywords
    text = """
    Machine learning is a field of artificial intelligence that uses statistical techniques
    to give computer systems the ability to learn from data without being explicitly programmed.
    """
keywords = textrank_keywords(text)
print("Extracted Keywords:", keywords)
```

Results:

To be attached with

```
Extracted Keywords: ['artificial', 'data', 'intelligence', 'learn', 'uses']
```

Observation:

In this experiment, the TextRank algorithm is used to extract important keywords from the given text by building a graph of words and applying the PageRank algorithm. It is observed that frequently occurring and strongly connected words receive higher scores and are selected as keywords. The method works well without requiring labeled data, but the results depend on text quality and preprocessing steps like stopword removal. Overall, TextRank effectively identifies the relevant terms from the document.

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Post Lab Exercise:

1. Write about "your ownself" (in not more than 500 words You know better about you, rather than ChatGPT!!). Extract top-25 keywords for your portfolio and analyze it in 3 categories: keyword can not give the real idea, keyword gives the real idea, keyword is irrelevant. Paste the code, your portfolio, output and your analysis.

Code:

```
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    scores = nx.pagerank(graph)

    ranked_words = sorted(scores.items(), key=lambda x: x[1], reverse=True)

    keywords = [word for word, score in ranked_words[:top_n]]
```

```
    return keywords
text = """
I am Mayur and I am pursuing a degree in Information and Communication Technology (ICT) in the Marwadi University. My origin is Wadhavan and Surendranagar, and I have great interests in software development and technology. I know such programming languages as C++, Python, and JavaScript and web development skills in HTML, CSS, and PHP. Some of the projects I have performed include event planning websites and tiffin service application. I am also knowledgeable in basic machine learning and data analysis. I am also familiar with microcontroller projects. I am a self-driven learner who would always desire to sharpen my skills and learn about new technologies.
"""

keywords = textrank_keywords(text)

print("Extracted Keywords:", keywords)
```

Output:

```
••• Extracted Keywords: ['skills', 'technology', 'also', 'development', 'projects']
```

Observation

The TextRank algorithm successfully extracts important keywords from the given paragraph by analyzing word relationships and co-occurrence. It builds a graph where words are connected based on proximity, and ranks them using the PageRank algorithm. The extracted keywords mainly include terms related to technology, programming, and skills, which accurately represent the core content of the text.

Analysis

The model effectively identifies meaningful words like “technology,” “development,” and “learning,” as they appear frequently and are strongly connected with other words. Removing stopwords and punctuation improves the quality of results by focusing only on relevant terms. However, the output may include some less meaningful or repeated words due to the limited context understanding. Overall, TextRank performs well for keyword extraction without needing labeled data, but its accuracy depends on text quality and preprocessing.